



Pneumonia in children: Epidemiology, pathogenesis, and etiology

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INTRODUCTION

Childhood pneumonia is an important cause of morbidity in resource-abundant countries, and morbidity and mortality in resource-limited countries. The epidemiology, microbiology, and pathogenesis of pneumonia in children will be reviewed here. The clinical features, diagnosis, and treatment of pneumonia in children and pneumonia in neonates (<28 days) are discussed separately:

- (See "[Community-acquired pneumonia in children: Clinical features and diagnosis](#)".)
- (See "[Community-acquired pneumonia in children: Outpatient treatment](#)".)
- (See "[Pneumonia in children >28 days old: Inpatient treatment](#)".)
- (See "[Neonatal pneumonia](#)".)

TERMINOLOGY

The terms pneumonia and pneumonitis strictly represent any inflammatory condition involving the lungs, which include the visceral pleura, connective tissue, airways, alveoli, and vascular structures.

Lower respiratory tract infection (LRTI) is frequently used interchangeably to include bronchitis, bronchiolitis, and pneumonia, or any combination of the three.

For this review, pneumonia will be defined as a condition typically associated with fever, respiratory symptoms, and evidence of parenchymal involvement, either by physical examination or the presence of infiltrates on chest radiography.

Bronchiolitis is discussed separately. (See "[Bronchiolitis in infants and children: Clinical features and diagnosis](#)", section on 'Clinical features'.)

EPIDEMIOLOGY

Incidence and hospitalization — The incidence of childhood pneumonia varies geographically.

- **Resource-abundant countries** – In resource-abundant countries, the annual incidence of pneumonia is estimated to be 3.3 per 1000 in children younger than 5 years and 1.45 per 1000 in children 0 to 16 years [1]. Approximately one-half of children younger than five years of age with community-acquired pneumonia require hospitalization [2]. Hospitalization rates for pneumonia (all causes) among children younger than two years in the United States decreased after introduction of the pneumococcal conjugate vaccine (PCV) to the routine childhood immunization schedule in 2000 (from 12 to 14 per 1000 population to 8 to 10 per 1000 population between 1997 and 2006) ([figure 1](#)) [3]. After licensure of 13-valent PCV in 2010, hospitalization rates for pneumonia (all causes) among children younger than two years in a single state declined to 4 per 1000, compared with 14 to 15 per 1000 before licensure of the 7-valent PCV (PCV7) and 8 to 9 per 1000 in the PCV7 years [4].
- **Resource-limited countries** – In a systematic review, the annual incidence of pneumonia in children younger than five years from resource-limited countries in 2015 was estimated to be 231 per 1000; 50 to 80 percent of children with severe pneumonia required hospitalization [2].

Mortality — In 2015, lower respiratory tract infections (LRTIs) accounted for nearly 800,000 deaths among children ≤19 years worldwide (31.1 per 100,000 population), second only to neonatal/preterm birth complications [5]. In observational studies in resource-abundant countries, the case fatality rate among hospitalized children <5 years of age was <1 percent [2,6]. In a systematic review, the case fatality rate among hospitalized children <5 years in resource-limited countries ranged from 0.3 to 15 percent [2].

Seasonality — Although both viral and bacterial pneumonia occur throughout the year, they are more prevalent during the colder months. The mechanisms responsible for this observed seasonality are likely multifactorial including environmental factors (eg, temperature, absolute humidity, sunlight) affecting both the pathogen (virus stability and transmissibility) as well as the host (eg, local, innate, and adaptive immune function) and human behavior patterns (indoor crowding during the winter months enhancing direct transmission of infected droplets) [7]. For reasons that are unknown, different viruses cause peaks of infection at different times during the respiratory virus season, and these peaks seldom occur simultaneously [8]. In tropical regions, peaks of infection follow no common pattern and can occur during either the wet or dry seasons.

Risk factors — Lower socioeconomic groups have a higher prevalence of LRTIs, which correlates best with family size, a reflection of environmental crowding. School-age children often introduce respiratory viral agents into households, resulting in secondary infections in their caregivers and siblings [8].

Underlying cardiopulmonary disorders and other medical conditions predispose to pneumonia and contribute to increasing severity. These include [9,10]:

- Congenital heart disease
- Bronchopulmonary dysplasia
- Cystic fibrosis
- Asthma
- Sickle cell disease
- Neuromuscular disorders, especially those associated with a depressed consciousness
- Some gastrointestinal disorders (eg, gastroesophageal reflux, tracheoesophageal fistula)
- Congenital and acquired immunodeficiency disorders

Cigarette smoke compromises natural pulmonary defense mechanisms by disrupting both mucociliary function and macrophage activity [11]. Exposure to cigarette smoke, especially if the mother smokes, increases the risk for pneumonia in infants younger than one year of age. (See "[Secondhand smoke exposure: Effects in children](#)".)

The use of cigarettes, alcohol, and other substances in adolescents may increase the risk of pneumonia by increasing the risk of aspiration through impairment of the cough and epiglottic reflexes. In addition, the use of alcohol has been associated with increased colonization of the oropharynx with aerobic gram-negative bacilli [12].

Effect of vaccines — Immunization with the *Haemophilus influenzae* type b (Hib) and pneumococcal conjugate vaccines protects children from invasive disease caused by these organisms. Hib was once a common cause of pneumonia in young children in the United States. However, it has been virtually eliminated as a result of routine immunization with Hib conjugate vaccines. (See "[Prevention of *Haemophilus influenzae* type b infection](#)", section on 'Efficacy/effectiveness'.)

The universal immunization of infants in the United States and other countries with the PCV has effectively decreased the incidence of pneumonia requiring hospitalization and other invasive *Streptococcus pneumoniae* infections in children [4,13-17]. (See "[Pneumococcal vaccination in children](#)", section on 'Efficacy and effectiveness'.)

Pneumococcal vaccination also reduces the risk of viral pneumonia. In a randomized trial, complete immunization with a 9-valent pneumococcal conjugate vaccine was associated with a 31 percent reduction (95% CI 15-43) in the incidence of pneumonia associated with any of seven respiratory viruses (influenza A/B, parainfluenza types 1 to 3, respiratory syncytial virus, adenovirus) in hospitalized children [18]. This observation suggests that the pneumonias associated with these viruses in hospitalized children are often because of concurrent pneumococcal infection.

PATHOGENESIS

Pneumonia occurs because of an impairment of host defenses, invasion by a virulent organism, and/or invasion by an overwhelming inoculum.

In the typical scenario, pneumonia follows an upper respiratory tract illness that permits invasion of the lower respiratory tract by bacteria, viruses, or other pathogens that trigger the immune response and produce inflammation [19,20]. The lower respiratory tract air spaces fill with white blood cells, fluid, and cellular debris. This process reduces lung compliance, increases resistance,

obstructs smaller airways, and may result in collapse of distal air spaces, air trapping, and altered ventilation-perfusion relationships [19]. Severe infection is associated with necrosis of bronchial or bronchiolar epithelium [21] and/or pulmonary parenchyma [22].

Acquisition — The agents that cause lower respiratory tract infection are most often transmitted by droplet spread resulting from close contact with a source case. Contact with contaminated fomites also may be important in the acquisition of viral agents, especially respiratory syncytial virus.

Most typical bacterial pneumonias (eg, *S. pneumoniae*) are the result of initial colonization of the nasopharynx followed by aspiration or inhalation of organisms. Invasive disease most commonly occurs upon acquisition of a new serotype of the organism with which the patient has not had previous experience, typically after an incubation period of one to three days. Occasionally, a primary bacteremia may precede the pneumonia. Atypical bacterial pathogens (eg, *Mycoplasma pneumoniae*, *Chlamydia pneumoniae*) attach to respiratory epithelial membranes through which they enter cells for replication.

The viral agents that cause pneumonia proliferate and spread by contiguity to involve lower and more distal portions of the respiratory tract.

Normal host defense — The pulmonary host defense system is complex and includes anatomic and mechanical barriers, humoral immunity, phagocytic activity, and cell-mediated immunity [23,24], as discussed below, with a focus on bacterial infection. The host response to respiratory viral infection is beyond the scope of this review; more information can be obtained from the reference [25].

- **Anatomic and mechanical barriers** – Anatomic and mechanical barriers in the upper airway form an important part of the host defense. Particles greater than 10 microns are efficiently filtered by the hairs in the anterior nares or are trapped on mucosal surfaces. The nasal mucosa contains ciliated epithelium and mucus-producing cells. The cilia beat synchronously, clearing the entrapped organisms through the nasopharynx via expulsion or swallowing. In the oropharynx, salivary flow, sloughing of epithelial cells, local production of complement and immunoglobulin (Ig)A, and bacterial interference from the resident flora serve as important factors in local host defense.

An intact epiglottic reflex helps to prevent aspiration of infected secretions, and the cough reflex helps to expel materials that may be aspirated. The sharp angles at which the central airways branch cause 5 to 10 micron particles to impact on mucosal surfaces, where they are entrapped in endobronchial mucus. Once entrapped, the ciliary system moves the particles upward out of the airways into the throat, where they are normally swallowed.

- **Humoral immunity** – Secretory IgA is the major immunoglobulin produced in the upper airways and accounts for 10 percent of the total protein concentration of nasal secretions. Although it is not a very good opsonizing agent, it does possess antibacterial and antiviral activity. IgG and IgM enter the airways and alveolar spaces predominantly via transudation from the blood and act to opsonize bacteria, activate complement, and neutralize toxin. Immunoglobulins, surfactant, fibronectin, and complement act as effective opsonins to help eliminate microorganisms (0.5 to 1 micron particles) that reach the terminal airways and alveoli. Free fatty acids, lysozyme, and iron-binding proteins also are present and may be microbicidal.
- **Phagocytic cells** – There are two populations of phagocytic cells in the lung: polymorphonuclear leukocytes from the blood and macrophages. There are several distinct populations of macrophages, which vary in their location and function:
 - The alveolar macrophage is located in the alveolar fluid and is the first phagocyte encountered by inert particles and potential pathogens entering the lung. If this cell is overwhelmed, it has the capacity to become a mediator of inflammation and produce cytokines that recruit neutrophils.
 - Interstitial macrophages are located in the lung connective tissue and serve both as phagocytic cells and antigen-processing cells.
 - The intravascular macrophage is located in capillary endothelial cells and phagocytizes and removes foreign material entering the lungs via the bloodstream.
- **Cell-mediated immunity** – Cell-mediated immunity is especially important against certain pathogens, including viruses and intracellular microorganisms that can survive within pulmonary macrophages. Although relatively few in number (5 to 10 percent of the total lung parenchyma cell population), lymphocytes play three critical roles: the production of antibody, cytotoxic activity, and the production of cytokines.

Pathologic patterns of pneumonia — There are five pathologic patterns of bacterial pneumonia [20]:

- **Lobar pneumonia** – Involvement of a single lobe or segment of a lobe. This is the classic pattern of *S. pneumoniae* pneumonia.
- **Bronchopneumonia** – Primary involvement of airways and surrounding interstitium. This pattern is sometimes seen in *Streptococcus pyogenes* and *Staphylococcus aureus* pneumonia.

- Necrotizing pneumonia (associated with aspiration pneumonia and pneumonia resulting from *S. pneumoniae* (image 1), *S. pyogenes*, and *S. aureus*).
- Caseating granuloma (as in pneumonia caused by *Mycobacterium tuberculosis* and the endemic mycoses).
- Interstitial and peribronchiolar with secondary parenchymal infiltration – This pattern typically occurs when a severe viral pneumonia is complicated by bacterial pneumonia.

There are two major pathologic patterns of viral pneumonia [20]:

- Interstitial pneumonia (image 2).
- Parenchymal infection.

ETIOLOGIC AGENTS

A large number of microorganisms have been implicated as etiologic agents of pneumonia in children (table 1 and table 2). The agents commonly responsible vary according to the age of the child and the setting in which the infection is acquired.

Community-acquired pneumonia — Community-acquired pneumonia (CAP) is an acute infection of the pulmonary parenchyma that is acquired in the community. The true prevalence of the various etiologic agents in CAP in children is difficult to determine. Studies investigating the etiology of childhood pneumonia have been performed in populations of various ages, in various settings, and using a variety of microbiologic techniques [6,26-37]. Because direct culture of infected lung tissue requires invasive techniques, published studies primarily use laboratory tests that provide indirect evidence of etiology (eg, nasopharyngeal culture, blood culture, polymerase chain reaction (PCR), serology). Interpretation of the results is further hampered by the failure to identify an organism in 15 to 35 percent of cases and the frequency of mixed infections (in 23 to 33 percent of cases) [1,6,38].

The most common causes of CAP in children vary with age.

Etiologic agents of pediatric pneumonia*

Microbial agent	Nonrespirable hosts
Bacteria	
<i>Chlamydia trachomatis</i>	First 3 months after birth
<i>Mycoplasma hominis</i>	First 3 months after birth
<i>Legionella pneumophila</i>	First 3 months after birth
<i>Campylobacter jejuni</i>	First 3 months after birth
<i>Haemophilus influenzae</i> type b	Unimmunized children <6 years
<i>Streptococcus pneumoniae</i>	Primarily children <6 years
<i>Streptococcus pyogenes</i>	Primarily children <6 years
<i>Chlamydia pneumoniae</i>	Primarily children <6 years
<i>Mycoplasma pneumoniae</i>	Primarily children <6 years
<i>Streptococcus pneumoniae</i>	All
<i>Mycobacterium tuberculosis</i>	All
Viruses	
<i>Respiratory syncytial virus</i>	Primarily children <6 years
Influenza A and B	Primarily children <6 years
Human metapneumovirus	Primarily children <6 years
<i>Parainfluenza</i> viruses 1, 2, and 3	Primarily children <6 years
Coronaviruses (229E, NL63, HKU1, SARS-CoV-2)	Primarily children <6 years (SARS-CoV-2 in all children)
Adenoviruses	Primarily children <6 years
Rhinoviruses	Primarily children <6 years
Cytomegalovirus	Primarily children <6 years
Fungi	
<i>Aspergillus fumigatus</i> (rare)	All
<i>Histoplasma capsulatum</i> , <i>Coccidioides immitis</i> , <i>Blastomyces dermatitidis</i>	

Etiologic agents of pediatric pneumonia*

Table 1 - larger image below

Less common etiologic agents of pediatric pneumonia*

Microbial agent	Nonrespirable hosts
Bacteria	
<i>Chlamydia trachomatis</i>	First 3 months after birth
<i>Mycoplasma hominis</i>	First 3 months after birth
<i>Legionella pneumophila</i>	First 3 months after birth
<i>Campylobacter jejuni</i>	First 3 months after birth
<i>Haemophilus influenzae</i> type b	Unimmunized children <6 years
<i>Streptococcus pneumoniae</i>	Primarily children <6 years
<i>Streptococcus pyogenes</i>	Primarily children <6 years
<i>Chlamydia pneumoniae</i>	Primarily children <6 years
<i>Mycoplasma pneumoniae</i>	Primarily children <6 years
<i>Streptococcus pneumoniae</i>	All
<i>Mycobacterium tuberculosis</i>	All
Viruses	
<i>Respiratory syncytial virus</i>	Primarily children <6 years
Influenza A and B	Primarily children <6 years
Human metapneumovirus	Primarily children <6 years
<i>Parainfluenza</i> viruses 1, 2, and 3	Primarily children <6 years
Coronaviruses (229E, NL63, HKU1, SARS-CoV-2)	Primarily children <6 years (SARS-CoV-2 in all children)
Adenoviruses	Primarily children <6 years
Rhinoviruses	Primarily children <6 years
Cytomegalovirus	Primarily children <6 years
Fungi	
<i>Aspergillus fumigatus</i> (rare)	All
<i>Histoplasma capsulatum</i> , <i>Coccidioides immitis</i> , <i>Blastomyces dermatitidis</i>	

Less common etiologic agents of pediatric pneumonia*

Table 2 - larger image below

In neonates — The etiology of pneumonia in neonates (infants <28 days of age) is discussed separately. (See "Neonatal pneumonia", section on 'Etiology'.)

In infants — Viruses are the most common cause of CAP in infants younger than one year. They account for >80 percent of CAP in children younger than two years [6]. Infants may also develop "afebrile pneumonia of infancy," a syndrome that typically occurs between two weeks and three to four months of age. It is classically caused by *Chlamydia trachomatis*, but other agents, such as cytomegalovirus (CMV), *Mycoplasma hominis*, and *Ureaplasma urealyticum*, also are implicated. (See "Chlamydia trachomatis infections in newborns and young infants", section on 'Pneumonia'.)

Infants with severe *Bordetella pertussis* infection also may develop pneumonia. (See "Pertussis infection in infants and children: Clinical features and diagnosis", section on 'Complications'.)

In children <5 years

- Viruses – Viruses are the most common etiology of CAP in older infants and children younger than five years of age [1,6,39]. They account for up to 50 percent of cases in young children [6].

Respiratory syncytial virus (RSV), a member of the *Pneumoviridae* virus family [40], is the most common viral pathogen responsible for pneumonia in children younger than five years [6,32,39,41,42]. RSV pneumonia frequently represents an extension of bronchiolitis. (See "Bronchiolitis in infants and children: Clinical features and diagnosis", section on 'Clinical features' and "Respiratory syncytial virus infection: Clinical features and diagnosis in infants and children", section on 'Clinical manifestations'.)

Other viral causes of pneumonia in children younger than five years, in decreasing order of likelihood, include [6,42]:

- Influenza A and B viruses.
- Human metapneumovirus is a common cause of lower respiratory tract infections (LRTIs) in children; most children have been infected by five years of age. (See ["Human metapneumovirus infections"](#).)
- A number of adenovirus serotypes (eg, 1, 2, 3, 4, 5, 7, 14, 21, and 35) have been reported to cause pneumonia; serotypes 3, 7, and 21 have been associated with severe and complicated pneumonia [43]. Adenovirus was found to be strongly associated with CAP in children younger than two years [42]. (See ["Pathogenesis, epidemiology, and clinical manifestations of adenovirus infections"](#), section on 'Clinical presentation'.)
- Parainfluenza viruses, usually type 3. (See ["Parainfluenza viruses in children"](#), section on 'Clinical presentation'.)
- Enterovirus D68 emerged as a significant pathogen of lower respiratory tract disease among American children in 2014 [44,45]. (See ["Enterovirus and parechovirus infections: Epidemiology and pathogenesis"](#), section on 'Periodicity and variability of disease by serotype' and ["Enterovirus and parechovirus infections: Clinical features, laboratory diagnosis, treatment, and prevention"](#), section on 'Respiratory disease'.)
- Coronaviruses (229E, OC43, NL63, HKU1) as well as MERS-CoV (responsible for Middle East respiratory syndrome) and severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2; responsible for coronavirus disease-2019 [COVID-19]) may also cause respiratory tract infections in children younger than five years [46-48]. However, their clinical impact has yet to be fully determined [41,42,49]. (See ["Common cold coronaviruses"](#), section on 'Respiratory syndromes' and ["Middle East respiratory syndrome coronavirus"](#), section on 'Clinical manifestations' and ["COVID-19: Clinical manifestations and diagnosis in children"](#).)
- Rhinovirus has been implicated as a cause of pneumonia using PCR assays on specimens from the upper respiratory tract [50,51], but its etiologic role is questioned [6,41,42,52,53], especially in children younger than five years [42].
- Human bocavirus and human parechovirus types 1, 2, and 3 also have been implicated as causes of LRTIs in children [54-57].
- Bacteria – Important bacterial causes of pneumonia in preschool children include *S. pneumoniae*, *H. influenzae* type b (Hib), nontypeable *H. influenzae*, *Moraxella catarrhalis*, *S. aureus*, *S. pyogenes*, and atypical bacteria. *S. pneumoniae*, *S. aureus*, and *S. pyogenes* are associated with increased morbidity and mortality [58-60].
 - *S. pneumoniae* is the most common typical bacterial pathogen causing pneumonia in all patients beyond the first few weeks after birth [10,61]. (See ["Pneumococcal pneumonia in children"](#), section on 'Epidemiology'.)
 - Hib is a rare cause of pneumonia in countries with universal childhood immunization.
 - *S. aureus* (particularly community-associated methicillin-resistant *S. aureus* [CA-MRSA]) and *S. pyogenes* are becoming increasingly frequent causes of CAP, particularly those complicated by necrosis and empyema [58,62]. In addition, these organisms occasionally cause pneumonia following influenza and chickenpox, respectively [59,63]. (See ["Epidemiology, clinical presentation, and evaluation of parapneumonic effusion and empyema in children"](#) and ["Clinical features of varicella-zoster virus infection: Chickenpox"](#).)

When associated with influenza, MRSA CAP can be particularly severe. (See ["Seasonal influenza in children: Clinical features and diagnosis"](#), section on '*S. pneumoniae*, *S. pyogenes*, or *S. aureus* coinfection'.)

- The prevalence of the atypical pathogens *M. pneumoniae* and *C. pneumoniae* may be increasing in preschool children with CAP [64]. (See ["Pneumonia caused by *Chlamydia pneumoniae* in children"](#) and ["Mycoplasma pneumoniae infection in children"](#), section on 'Epidemiology'.)

In children ≥5 years

- *S. pneumoniae* is the most common typical bacterial cause of pneumonia in children older than five years (see ["Pneumococcal pneumonia in children"](#), section on 'Epidemiology')
- *M. pneumoniae* is more common among children ≥5 years than among younger children [6,65] (see ["Mycoplasma pneumoniae infection in children"](#), section on 'Epidemiology')
- *C. pneumoniae* also is emerging as a frequent cause of pneumonia in older children and young adults [66] (see ["Pneumonia caused by *Chlamydia pneumoniae* in children"](#))
- Although viruses primarily cause pneumonia in young children, the COVID-19 pandemic has demonstrated that SARS-CoV-2 can be responsible for severe pneumonia in older children/adolescents who have risk factors such as obesity (see ["COVID-19: Clinical manifestations and diagnosis in children"](#))

In areas where CA-MRSA is prevalent, CA-MRSA is an important cause of CAP complicated by empyema and necrosis [58,67]. When associated with influenza, MRSA CAP can be particularly severe [59,63]. (See "[Epidemiology, clinical presentation, and evaluation of parapneumonic effusion and empyema in children](#)" and "[Methicillin-resistant *Staphylococcus aureus* \(MRSA\) in children: Epidemiology and clinical spectrum](#)", section on 'Epidemiology and risk factors'.)

Aspiration pneumonia — When there is a predisposition to aspiration, pneumonia may be caused by anaerobic oral flora, including:

- Anaerobic streptococci (eg, *Peptostreptococcus*)
- *Fusobacterium* spp
- *Bacteroides* spp
- *Prevotella melaninogenica*

Risk factors for aspiration include a history of seizure, anesthesia, or other episodes of reduced level of consciousness, neurologic disease, dysphagia, gastroesophageal reflux, alcohol or substance use disorders, use of a nasogastric tube, or foreign body aspiration.

Hospital-acquired pneumonia — Hospital-acquired pneumonia occurs ≥ 48 hours after admission and does not appear to be incubating at the time of admission. Hospital-acquired bacterial pneumonia is usually caused by gram-negative bacilli or *S. aureus*. Hospital-acquired pneumonia frequently occurs in intensive care units where mechanical ventilation, indwelling catheters, and administration of broad-spectrum antibiotics are common. (See "[Pneumonia in children >28 days old: Inpatient treatment](#)".)

In addition, during the winter respiratory viral season, hospitalized children are at risk for hospital-acquired pneumonia caused by RSV, parainfluenza, and influenza viruses. (See "[Seasonal influenza in children: Clinical features and diagnosis](#)" and "[Parainfluenza viruses in children](#)", section on 'Clinical presentation' and "[Respiratory syncytial virus infection: Clinical features and diagnosis in infants and children](#)", section on 'Transmission and incubation period'.)

Special populations

Immunocompromised — The causes of pneumonia in immunocompromised hosts include all of the pathogens mentioned above, as well as a variety of other organisms, as discussed below.

Gram-negative bacilli and *S. aureus* are common etiologies in neutropenic patients or in those with white blood cell defects. Clinically significant legionellosis usually is seen only in immunocompromised hosts with an exposure to an aquatic reservoir of *Legionella pneumophila*, such as a river, lake, air-conditioning cooling tower, or water distribution systems [68,69]. However, seroepidemiologic studies suggest that subclinical or minor infections occur in children [70,71]. (See "[Microbiology, epidemiology, and pathogenesis of *Legionella* infection](#)".)

Opportunistic fungi, such as *Aspergillus* spp, *Mucoraceae* spp, and *Fusarium* spp, also are a concern in neutropenic patients and in those receiving immunosuppressive therapies that impair the cell-mediated response. One of the more common pneumonia pathogens diagnosed in human immunodeficiency virus (HIV)-infected patients is *Pneumocystis jirovecii*, which was formerly called *Pneumocystis carinii* [72]. (See "[Epidemiology and clinical manifestations of invasive aspergillosis](#)" and "[Mycology, pathogenesis, and epidemiology of *Fusarium* infection](#)" and "[Pediatric HIV infection: Epidemiology, clinical manifestations, and outcome](#)", section on 'Pneumocystis jirovecii pneumonia'.)

Viral causes of pneumonia, which may be life-threatening in the immunocompromised host, including the post-solid organ and stem cell transplant populations, include:

- Common community-acquired viral agents such as [73,74]:
 - RSV (see "[Respiratory syncytial virus infection: Clinical features and diagnosis in infants and children](#)")
 - Adenovirus (see "[Pathogenesis, epidemiology, and clinical manifestations of adenovirus infections](#)")
 - Influenza (see "[Seasonal influenza in children: Clinical features and diagnosis](#)")
 - Parainfluenza (see "[Parainfluenza viruses in children](#)", section on 'Risk and protective factors')
 - Rhinovirus (see "[Epidemiology, clinical manifestations, and pathogenesis of rhinovirus infections](#)")
 - Human metapneumovirus (see "[Human metapneumovirus infections](#)")
- SARS-CoV-2. (See "[COVID-19: Clinical manifestations and diagnosis in children](#)".)

- Rubeola (Hecht giant-cell pneumonia). (See "[Measles: Clinical manifestations, diagnosis, treatment, and prevention](#)", section on '[Immunocompromised patients](#)'.)
- Varicella-zoster virus (VZV). (See "[Clinical features of varicella-zoster virus infection: Chickenpox](#)", section on '[Pneumonia](#)'.)
- CMV. (See "[Overview of cytomegalovirus \(CMV\) infections in children](#)", section on '[CMV infections in immunocompromised hosts](#)'.)
- Epstein-Barr virus, which may be the trigger for lymphoid interstitial pneumonitis (LIP), an indolent but progressive process that occurs in children infected with HIV. LIP can also be seen in patients with common variable immunodeficiency. (See "[Clinical manifestations and treatment of Epstein-Barr virus infection](#)" and "[Classification of diffuse lung disease \(interstitial lung disease\) in infants and children](#)", section on '[Disorders of the immunocompromised host](#)'.)

Cystic fibrosis — Young children with cystic fibrosis frequently are infected with *S. aureus*, *Pseudomonas aeruginosa*, and *H. influenzae* (mostly nontypeable strains). Later in the course of the disease, multiple drug-resistant gram-negative organisms, such as *Burkholderia cepacia*, *Stenotrophomonas maltophilia*, and *Achromobacter xylosoxidans*, can be recovered. *Aspergillus* spp and nontuberculous mycobacteria also may cause disease in this population. Cystic fibrosis lung disease is discussed in detail separately. (See "[Cystic fibrosis: Clinical manifestations of pulmonary disease](#)" and "[Cystic fibrosis: Overview of the treatment of lung disease](#)" and "[Cystic fibrosis: Antibiotic therapy for chronic pulmonary infection](#)".)

Sickle cell disease — The prevalence of pneumonia is increased in children with sickle cell disease [75]. Atypical bacterial pathogens (eg, *M. pneumoniae*, *C. pneumoniae*) appear to be most frequent and are more commonly associated with the acute chest syndrome. Other bacterial causes of pneumonia in children with sickle cell disease include *S. pneumoniae* associated with functional asplenia, *S. aureus*, and *H. influenzae* [10]. (See "[Acute chest syndrome \(ACS\) in sickle cell disease \(adults and children\)](#)", section on '[Common triggers](#)'.)

Environmental considerations

Geography — Residence in or travel to specific geographic areas should suggest endemic pathogens:

- Tuberculosis is most common in immigrants from countries with a high prevalence of infection (eg, countries throughout Asia, Africa, Latin America, and eastern Europe) (📄 figure 2). (See "[Epidemiology of tuberculosis](#)".)
- Measles pneumonia is common in the resource-limited countries. (See "[Measles: Clinical manifestations, diagnosis, treatment, and prevention](#)".)
- *Coccidioides immitis* is endemic to the southwestern United States, northern Mexico, and parts of Central and South America. (See "[Primary pulmonary coccidioidal infection](#)".)
- *Blastomyces dermatitidis*, causing blastomycosis, is endemic in the southeastern and central United States and the midwestern states bordering the Great Lakes. (See "[Mycology, pathogenesis, and epidemiology of blastomycosis](#)" and "[Treatment of blastomycosis](#)".)
- In the United States, *Histoplasma capsulatum* is most common in the Ohio, Mississippi, and Missouri River valleys but has been identified in all regions [76,77]. It also occurs in Canada, Central America, eastern and southern Europe, parts of Africa, eastern Asia, and Australia.

Activities potentially leading to exposure to bird droppings and bat guano may be suggestive [78]. These include gardening, construction, cleaning of barns and outbuildings, and spelunking. (See "[Pathogenesis and clinical features of pulmonary histoplasmosis](#)" and "[Diagnosis and treatment of pulmonary histoplasmosis](#)".)

- In the United States, hantavirus cardiopulmonary syndrome (acute febrile illness associated with respiratory failure, shock, and high mortality) occurs predominantly west of the Mississippi River (in the "four corners" region of the United States where the borders of Colorado, New Mexico, Arizona, and Utah meet) after environmental exposure to infected deer mouse (*Peromyscus maniculatus*) saliva, urine, or feces. Activities associated with exposure include cleaning of barns and outbuildings, trapping rodents, animal herding, and farming with hand tools. (See "[Epidemiology and diagnosis of hantavirus infections](#)" and "[Hantavirus cardiopulmonary syndrome](#)".)
- MERS is endemic in countries in or near the Arabian Peninsula. (See "[Middle East respiratory syndrome coronavirus](#)".)

Animal exposures — Histoplasmosis is associated with exposure to bird droppings and bat guano, and hantavirus infection is associated with exposure to an infected deer mouse. Other causes of pneumonia that are associated with animal exposure include:

- *Chlamydia psittaci* (psittacosis), which is transmitted to humans predominantly from domestic and wild birds. (See "[Psittacosis](#)".)

- *Coxiella burnetii* (Q fever), which is associated with exposure to parturient sheep, goats, cattle, and cats (or exposure to dust/soil contaminated by these animals). Additional information about Q fever is available on the [United States Centers for Disease Control and Prevention website \[79,80\]](#). (See "[Q fever: Epidemiology, microbiology, and diagnostic tests](#)", section on 'Animal reservoirs'.)

Other exposures — Exposure to individuals at high risk for tuberculosis is a risk factor for the development of tuberculosis in children. High-risk individuals include people experiencing homelessness, recent immigrants from endemic regions ([figure 2](#)), incarcerated individuals, and HIV-infected patients. (See "[Epidemiology of tuberculosis](#)".)

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See "[Society guideline links: Pediatric pneumonia](#)".)

INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5th to 6th grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more sophisticated, and more detailed. These articles are written at the 10th to 12th grade reading level and are best for patients who want in-depth information and are comfortable with some medical jargon.

Here are the patient education articles that are relevant to this topic. We encourage you to print or email these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient education" and the keyword[s] of interest.)

- Basics topic (see "[Patient education: Pneumonia in children \(The Basics\)](#)")

SUMMARY

- **Epidemiology** – Pneumonia is more common in children younger than five years of age than in older children and adolescents. Risk factors for pneumonia include environmental crowding, having school-age siblings, and underlying cardiopulmonary and other medical disorders. (See "[Epidemiology](#)" above.)
- **Etiology** – Pneumonia can be caused by a large number of microorganisms ([table 1](#) and [table 2](#)). The agents commonly responsible vary according to the age of the child and the setting in which the infection is acquired. (See "[Etiologic agents](#)" above.)
 - Community-acquired pneumonia
 - Children <5 years – Viruses are most common. However, bacterial pathogens, including *Streptococcus pneumoniae*, *Staphylococcus aureus*, and *Streptococcus pyogenes*, also are important. (See "[In children <5 years](#)" above.)
 - Otherwise healthy children ≥5 years – *S. pneumoniae*, *Mycoplasma pneumoniae*, and *Chlamydia pneumoniae* are most common. (See "[In children ≥5 years](#)" above.)
 - Children of all ages – Community-associated methicillin-resistant *S. aureus* is an increasingly important pathogen.
 - Necrotizing pneumonia – Common causes of necrotizing pneumonia include *S. pneumoniae*, *S. aureus*, and *S. pyogenes*. (See "[Pathologic patterns of pneumonia](#)" above.)
 - Aspiration pneumonia – Aspiration pneumonia is usually caused by anaerobic oral flora. (See "[Aspiration pneumonia](#)" above.)
 - Hospital-acquired pneumonia – Hospital-acquired pneumonia is usually caused by gram-negative bacilli or *S. aureus*. (See "[Hospital-acquired pneumonia](#)" above.)

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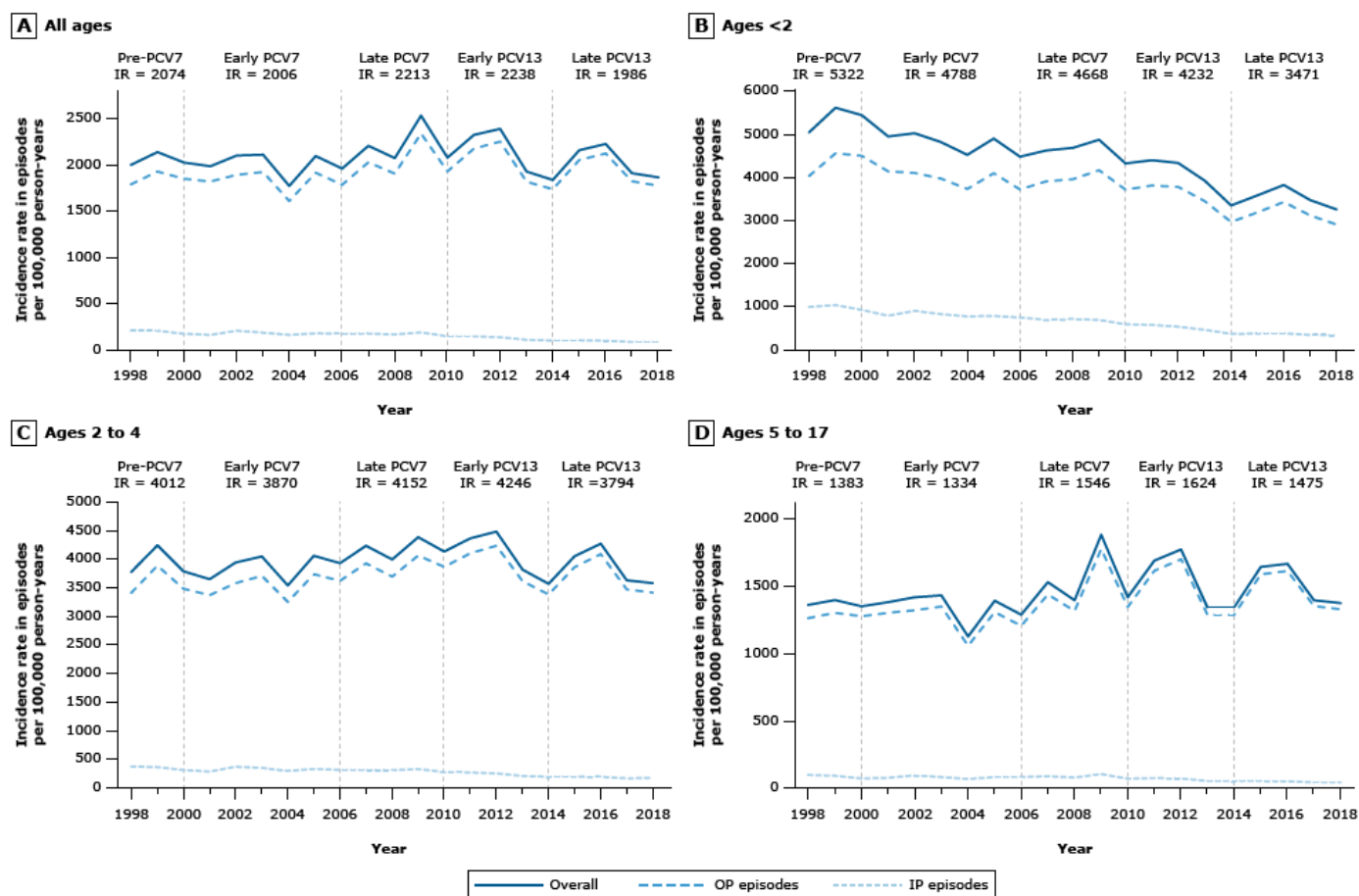
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GRAPHICS

Figure 1: Trends in annual incidence rates of all-cause pneumonia episodes per 100,000 person-years among commercially insured children aged <18 years, 1998-2018



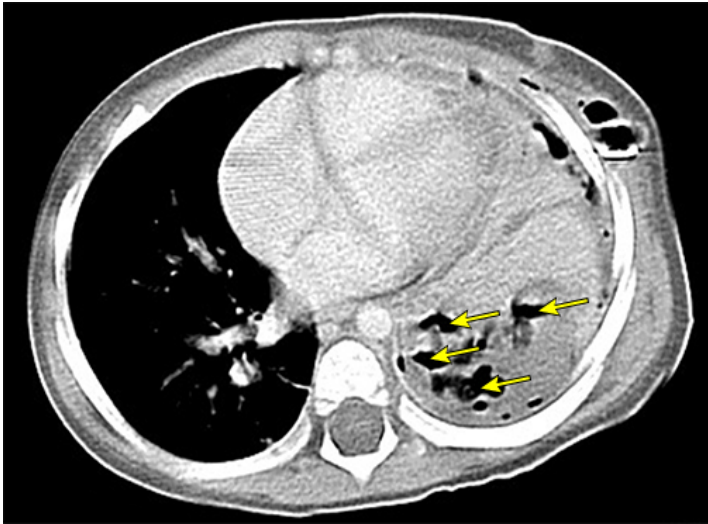
Average incidence rates for total all-cause pneumonia episodes are shown for each pneumococcal conjugate vaccine period. All patients' month and day of birth was imputed as July 1st. Age at onset was calculated as the difference between condition start date and imputed birth date. Patients with negative age at onset were included in the age 0 to 1 year cohort.

IP: inpatient; IR: incidence rate; OP: outpatient; PCV: pneumococcal conjugate vaccine.

From: Hu T, Sarpong EM, Song Y, et al. Incidence of non-invasive all-cause pneumonia in children in the United States before and after the introduction of pneumococcal conjugate vaccines: A retrospective claims database analysis. *Pneumonia (Nathan)* 2023; 15:8. Copyright © 2023 Merck & Co., Inc., Rahway, NJ, USA and its affiliates. <https://pneumonia.biomedcentral.com/articles/10.1186/s41479-023-00109-5> (Accessed on June 18, 2024). Reproduced under the terms of the [Creative Commons Attribution License 4.0](#).

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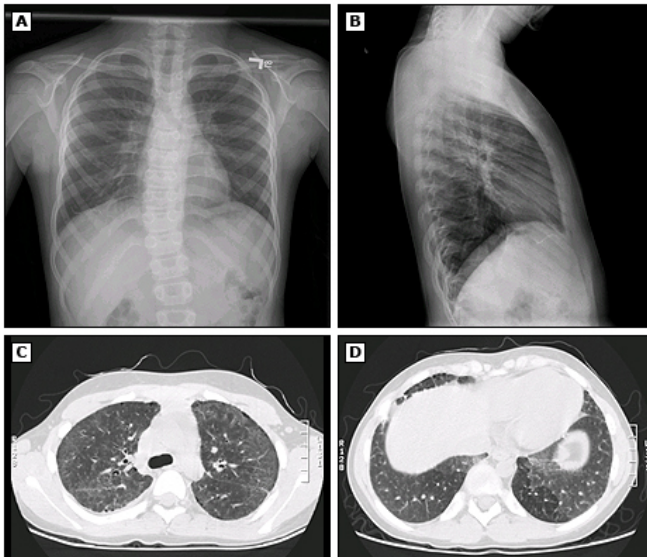
Image 1: Computed tomography: Left-sided *Streptococcus pneumoniae* necrotizing pneumonia



Computed tomography of the chest demonstrating left-sided necrotizing pneumonia. Note the consolidation of the left lung parenchyma with focal areas of liquefaction (arrows).

Courtesy William J Barson, MD.

Graphic 86362 Version 3.0

Image 2: Lymphocytic interstitial pneumonitis

(A, B) In a 17-year-old with non-HIV-related LIP, frontal and lateral chest radiographs show bilateral ground glass abnormality.
(C, D) In the same patient, HRCT scans at 2 levels confirm widespread ground glass abnormality, with mosaic pattern, and some subpleural honeycomb cysts.

LIP: lymphocytic interstitial pneumonitis; HRCT: high-resolution computed tomography.

Courtesy of Cynthia Epstein, MD and Leland L Fan, MD.

Graphic 82288 Version 6.0

*

Table 1: Etiologic agents of pediatric pneumonia

Microbial agent	Susceptible hosts
Bacteria	
<i>Chlamydia trachomatis</i>	First 3 months after birth
<i>Mycoplasma hominis</i>	First 3 months after birth
<i>Treponema pallidum</i>	First 3 months after birth
<i>Ureaplasma urealyticum</i>	First 3 months after birth
<i>Haemophilus influenzae</i> type b	Unimmunized children <4 years
<i>Staphylococcus aureus</i>	Primarily children <5 years
<i>Streptococcus pyogenes</i>	Primarily children <5 years
<i>Chlamydia pneumoniae</i>	Primarily children ≥5 years
<i>Mycoplasma pneumoniae</i>	Primarily children ≥5 years
<i>Streptococcus pneumoniae</i>	All
<i>Mycobacterium tuberculosis</i>	All
Viruses	
Respiratory syncytial virus	Primarily children <5 years
Influenza A and B	Primarily children <5 years
Human metapneumovirus	Primarily children <5 years
Parainfluenza viruses 1, 2, and 3	Primarily children <5 years
Coronaviruses (229E C43, NL63, HKU1, SARS-CoV-2)	Primarily children <5 years (SARS-CoV-2: all children)
Adenoviruses	Primarily children <5 years
Rhinovirus	Primarily children <5 years
Cytomegalovirus	Primarily children <1 year
Fungi	
Endemic mycoses (eg, histoplasmosis, coccidioidomycosis, blastomycosis)	All

SARS-CoV-2: severe acute respiratory syndrome coronavirus 2.

* Excluding neonatal pneumonia (in infants <28 days of age).

Data from:

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Graphic 54228 Version 18.0

Table 2: Less common etiologic agents of pediatric pneumonia*

Microbial agent	Susceptible hosts
Bacteria	
<i>Haemophilus influenzae</i> (typeable and nontypeable)	Unimmunized (<i>H. influenzae</i> type b)
<i>Moraxella catarrhalis</i>	
<i>Neisseria meningitidis</i> (often group Y)	
<i>Klebsiella</i> spp	Immunocompromised hosts, nosocomial pathogen
<i>Escherichia coli</i>	Immunocompromised hosts, nosocomial pathogen
<i>Enterobacter</i> spp	Immunocompromised hosts, nosocomial pathogen
<i>Pseudomonas aeruginosa</i>	Immunocompromised hosts, nosocomial pathogen, cystic fibrosis
<i>Burkholderia cepacia</i>	Cystic fibrosis patients
<i>Achromobacter xylosoxidans</i>	Cystic fibrosis patients, nosocomial pathogen
<i>Stenotrophomonas maltophilia</i>	Cystic fibrosis patients, nosocomial pathogen
<i>Legionella pneumophila</i>	Immunocompromised hosts
<i>Pseudomonas pseudomallei</i>	Travelers to or residents of endemic areas
<i>Francisella tularensis</i>	Exposure to a particular animal reservoir (rabbits) or insect vector; bioterrorist activity
<i>Brucella abortus</i>	Exposure to a particular animal reservoir (cattle, goats)
<i>Chlamydia psittaci</i>	Exposure to a particular animal reservoir (psittacines – parrots, parakeets, macaws, and virtually all domestic and wild birds)
<i>Leptospira</i> spp	Exposure to a particular animal reservoir
<i>Yersinia pestis</i>	Exposure to a particular animal reservoir or insect vector (rats); bioterrorist activity
Anaerobic mouth flora (<i>Prevotella</i> spp, <i>Fusobacterium</i> , <i>Bacteroides</i> spp)	Aspiration
<i>Mycobacterium tuberculosis</i>	Travelers to or residents of endemic areas, contact with known TB, homeless, recent immigrants from endemic areas, prisoners, and HIV-infected individuals
Nontuberculous mycobacterium	Cystic fibrosis patients, immunocompromised hosts
<i>Bordetella pertussis</i>	Infants, exposure to adult with a cough illness
Fungi	
<i>Coccidioides immitis</i>	Travelers to or residents of endemic areas (southwest United States)
<i>Histoplasma capsulatum</i>	Travelers to or residents of endemic areas (Ohio, Mississippi, and Missouri River valleys); also occurs in Canada, Central America, eastern and southern Europe, parts of Africa, eastern Asia, and Australia
<i>Blastomyces dermatitidis</i>	Travelers to or residents of endemic areas (southeastern and central United States and midwestern states bordering the Great Lakes)
<i>Candida</i> spp	Nosocomial pathogen, immunocompromised hosts, cystic fibrosis
<i>Aspergillus</i> spp	Nosocomial pathogen, immunocompromised hosts, cystic fibrosis
<i>Pneumocystis jiroveci</i> (formerly <i>carinii</i>)	Immunocompromised hosts
Rickettsiae	
<i>Coxiella burnetii</i>	Exposure to a particular animal reservoir (sheep)
<i>Rickettsia rickettsii</i>	Exposure to a particular insect vector
Viruses	
Rubeola	Travelers to or residents of endemic areas
Herpes simplex virus	Neonates and immunocompromised hosts
Cytomegalovirus	Immunocompromised hosts, first three months after birth
Epstein-Barr virus	Immunocompromised hosts
Varicella zoster virus	Immunocompromised hosts
Coronavirus (SARS-CoV, MERS-CoV, SARS-CoV-2)	

Enterovirus (D68)	Primarily children <5 years, adolescents, and children with asthma
Rubella	
Hantavirus	Travelers to or residents of endemic areas, exposure to a particular animal reservoir (mouse droppings)
HIV	
Mumps	
Human bocavirus	Primarily children <5 years
Parechovirus	Primarily children <5 years
Avian influenza	Exposure to birds; travel to affected area

TB: tuberculosis; HIV: human immunodeficiency virus; SARS: severe acute respiratory syndrome; CoV: coronavirus; MERS: Middle East respiratory syndrome.

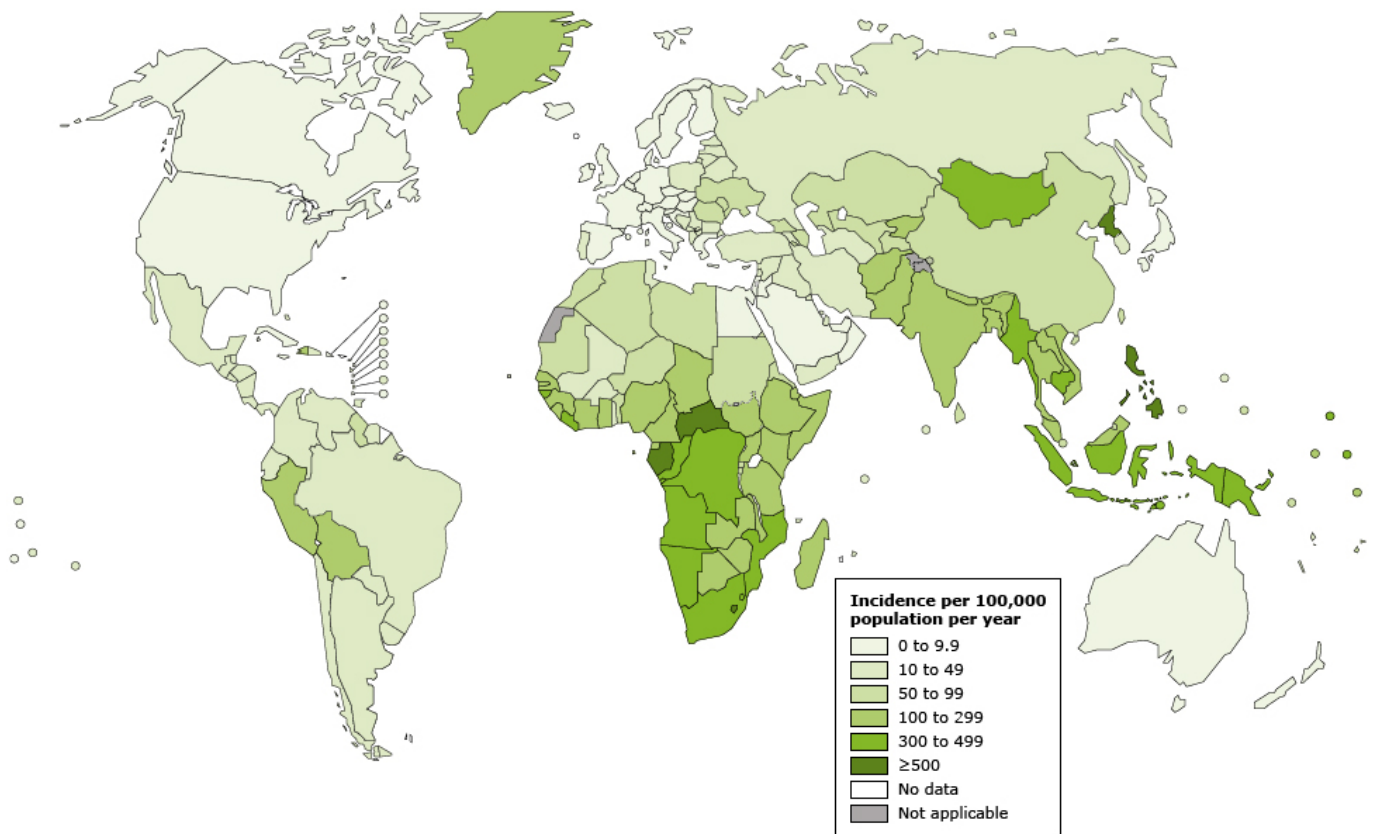
* Excluding neonatal pneumonia (in infants <28 days of age).

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Graphic 71263 Version 21.0

Figure 2: Estimated tuberculosis incidence rates, 2022



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